Semester 5 · Common Course · No Credit · 60 periods

Industrial Management and Safety

Complete handbook for management principles, HR, ownership, wages, quality, ISO, TQM, stores, CPM, PERT, LPP, transportation problem and industrial safety.

Course purpose

This subject gives technicians the management language needed inside industry: planning, organizing, staffing, quality systems, stores, project scheduling, quantitative decisions and safety discipline.

Malayalam: Industry-ൽ technical skill മാത്രം മതി എന്ന ചിന്ത തെറ്റാണ്. manpower, material, quality, project time, safety എന്നിവ manage ചെയ്യാൻ അറിയണം.

Official details

Item	Value
Program	Diploma in Engineering and Technology / Commercial Practice / Management
Course code	5001
Title	Industrial management and safety
Category	Common Course
Periods	4 per week, 60 per semester

CO1

Management principles, approaches, HRM, organization, wages and incentive plans.

CO2

Quality, ISO, TQM, purchase, inventory, stores and modern management concepts.

CO3 + CO4

CPM, PERT, LPP, transportation and industrial safety, disputes and settlement.

Roadmap

Module 1 · CO1 · 15 Hours

Principles and Functions of Management

Management fundamentals

Management is the process of planning, organizing, staffing, directing and controlling resources to achieve objectives. Administration frames policies; management executes them. Taylor focused scientific management, time study, standardization and productivity. Fayol gave administrative principles such as division of work, authority, discipline, unity of command and scalar chain.

HR and organization

Line, functional and line-staff organization structures decide authority flow. HRM covers manpower planning, recruitment, training, job evaluation, merit rating and performance appraisal. Ownership forms include sole proprietorship, partnership, private and public limited companies, cooperative societies and transnational organizations.

Wages and incentives

Wages may be time-rate, piece-rate or incentive-based. Halsey and Rowan plans reward workers for time saved while controlling excessive incentive payout.

Halsey bonus = 50 percent of time saved × hourly rate.

Rowan bonus = time taken × hourly rate × time saved / time allowed.

Expected questions

1. Compare management and administration.
2. Explain Taylor and Fayol principles.
3. Explain job evaluation, merit rating and training methods.
4. Solve a simple Halsey or Rowan wage problem.

Module 2 · CO2 · 13 Hours

Quality and Material Management

Quality, ISO and TQM

Quality means fitness for use and conformance to requirement. ISO 9000 gives documented quality management requirements. Implementation includes preparatory study, documentation, implementation, audit, registration and certification. TQM links every department to mission, vision and quality policy.

Purchase and stores

Purchase department obtains right material at right quality, quantity, price and time. Tenders may be single or open. Stores management controls receipt, inspection, bin cards, store ledger, issue and stock verification.

Inventory and modern concepts

Inventory management balances holding cost and shortage risk. EOQ and ABC analysis are common. Modern concepts include JIT, quality circle, zero defect, 5S, MIS and lean management.

Expected questions

1. Explain ISO 9000 implementation steps.
2. Write link between ISO and TQM.
3. Explain purchase procedure and store records.
4. Explain EOQ, ABC, JIT and 5S briefly.

Module 3 · CO3 · 20 Hours

Project and Quantitative Techniques

CPM and PERT

Network analysis plans project activities, sequence and time. CPM uses deterministic activity times and identifies critical path. PERT uses optimistic, most likely and pessimistic times to calculate expected time.

PERT expected time = $(a + 4m + b) / 6$.

LPP and transportation

Linear programming optimizes an objective function subject to constraints. Graphical method is used for two variables. Transportation problem finds initial feasible solution using North-West corner rule.

Worked example

If PERT values are $a=2$, $m=5$, $b=8$ days, expected time = $(2+20+8)/6 = 5$ days.

Expected time = 5 days.

Expected questions

1. Define EFT, LFT, float, dummy activity and critical path.
2. Compare CPM and PERT.
3. Formulate an LPP from given data.
4. Find initial feasible solution by North-West corner rule.

Module 4 · CO4 · 10 Hours

Industrial Safety and Disputes

Safety terms

Accident is an unwanted event causing injury or damage. Incident is a near miss or event that could cause loss. Safety performance uses frequency rate, severity rate and incidence rate. Causes include mechanical, environmental and personal factors.

Accident prevention

The 4E method covers Engineering, Education, Enforcement and Enthusiasm. Management, safety officers, safety council and government norms are all responsible for safe work.

Malayalam: Safety PPE മാത്രം അല്ല. Machine guard, training, rules, housekeeping, inspection, permit system എന്നിവ ചേർന്നതാണ്.

Hazards and disputes

Precautions are required for toxic, flammable, electrical shock and material handling situations. Industrial disputes may be settled by collective bargaining, conciliation, mediation and arbitration.

Expected questions

1. Define accident, incident, unsafe act and accident proneness.
2. Explain 4Es of accident prevention.
3. Explain safety precautions in electrical shock and material handling.

4. Explain settlement of industrial disputes.

Formula / Rule Bank

PERT expected time = (a+4m+b)/6. Wage bonus depends on time saved. Total project duration is length of critical path. Safety improvement must remove hazard first, not only warn workers.

Model question paper

1. Explain principles and functions of management.
2. Explain ISO, TQM, purchase and stores management.
3. Solve CPM/PERT and formulate LPP from data.
4. Explain industrial safety, 4E prevention and dispute settlement.

MCQ practice

1. ISO mainly relates to quality system. 2. Critical path has zero float. 3. 5S improves workplace organization. 4. Arbitration is a dispute settlement method.

Complete answer guide

Write answers in engineering format: definition, purpose, steps, diagram/table if useful, example and conclusion. For numerical problems show given data, formula, substitution and final unit. For safety answers mention hazard source, preventive engineering control, training, PPE, inspection and statutory rule.

References: O.P. Khanna, N.D. Vohra, L.S. Srinath, Panneerselvam, Shailendra Kale, L.M. Deshmukh, NPTEL and SWAYAM resources.